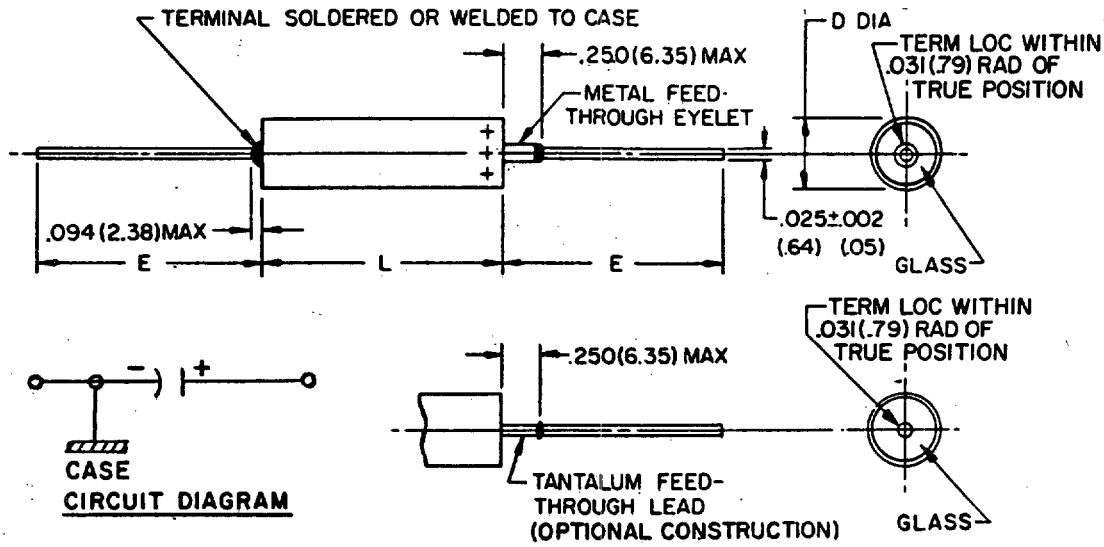
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG) - 85°C (VOLTAGE DERATED TO 125°C), STYLES CL66 AND CL67

INACTIVE FOR NEW EQUIPMENT DESIGN AFTER DATE OF THIS SPECIFICATION; USE MIL-C-39006/9

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



Case size	Dimensions			
	1/ L	D		E
	+.031 (.79) -.016 (.41)	±.016 (.41) (CL66)	Max (CL67)	± .250 (6.35)
T1	.453 (11.51)	.188 (4.78)	.219 (5.56)	1.500 (38.10)
T2	.641 (16.28)	.281 (7.14)	.312 (7.92)	2.250 (57.15)
T3	.766 (19.46)	.375 (9.53)	.406 (10.31)	2.250 (57.15)

1/ For style CL67, nonshrinkable sleeving shall extend .016 inch minimum, .062 inch maximum, beyond each end of the case; shrinkable sleeving shall lap over the ends of the case.

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Styles CL66 and CL67 capacitors.

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation 1/	Voltage			Capacitance 2/	Cap. tolerance	DC leakage (max)		Dissipation factor (max)	Impedance (max)	Capacitance change			Case size
	Rated (+85°C)	Derated (+125°C)	Surge (+85°C)			+25°C	+85°C & +125°C			-55°C	+85°C	+125°C	
	volts, dc	volts, dc	volts, dc	µF		µA	µA	percent	ohms	percent	percent	percent	
CL6-BB300-PG	6	4	6.9	30.0	J, K, M	1	2	9.1	100	-40	+10.5	+12	T1
CL6-BB680-PG	6	4	6.9	68.0*	J, K, M	1	2	20.4	60	-40	+14	+16	T1
CL6-BB141-PG	6	4	6.9	140.0	J, K, M	1	3	21.3	40	-40	+14	+16	T2
CL6-BB271-PG	6	4	6.9	270.0*	J, K, M	1	7	81.8	25	-44	+17.5	+20	T2
CL6-BB331-PG	6	4	6.9	330.0	J, K, M	2	8	49.6	20	-44	+14	+16	T3
CL6-BB561-PG	6	4	6.9	560.0*	J, K, M	2	13	128.0	25	-64	+17.5	+20	T3
CL6-BC250-PG	8	5	9.2	25.0	J, K, M	1	2	7.6	100	-40	+10.5	+12	T1
CL6-BC560-PG	8	5	9.2	56.0*	J, K, M	1	2	17.0	59	-40	+14	+16	T1
CL6-BC221-PG	8	5	9.2	220.0*	J, K, M	1	7	66.4	30	-44	+17.5	+20	T2
CL6-BC431-PG	8	5	9.2	430.0*	J, K, M	2	14	91.5	25	-64	+17.5	+20	T3
CL6-BD200-PG	10	7	11.5	20.0	J, K, M	1	2	6.1	175	-32	+10.5	+12	T1
CL6-BD470-PG	10	7	11.5	47.0*	J, K, M	1	2	18.1	100	-36	+14	+16	T1
CL6-BD101-PG	10	7	11.5	100.0	J, K, M	1	4	15.2	60	-36	+14	+16	T2
CL6-BD181-PG	10	7	11.5	180.0*	J, K, M	1	7	54.4	40	-36	+14	+16	T2
CL6-BD251-PG	10	7	11.5	250.0	J, K, M	2	10	37.8	30	-40	+14	+16	T3
CL6-BD391-PG	10	7	11.5	390.0*	J, K, M	2	16	87.6	25	-64	+17.5	+20	T3
CL6-BE150-PG	15	10	17.2	15.0	J, K, M	1	2	5.7	155	-24	+10.5	+12	T1
CL6-BE330-PG	15	10	17.2	33.0*	J, K, M	1	2	12.5	90	-28	+14	+16	T1
CL6-BE700-PG	15	10	17.2	70.0	J, K, M	1	4	13.1	75	-28	+14	+16	T2
CL6-BE121-PG	15	10	17.2	120.0*	J, K, M	1	7	36.8	50	-28	+17.5	+20	T2
CL6-BE171-PG	15	10	17.2	170.0	J, K, M	2	10	25.4	35	-32	+14	+16	T3
CL6-BE271-PG	15	10	17.2	270.0*	J, K, M	2	16	60.9	30	-56	+17.5	+20	T3
CL6-BG100-PG	25	15	28.8	10.0	J, K, M	1	2	4.6	220	-16	+8	+9	T1
CL6-BG220-PG	25	15	28.8	22.0*	J, K, M	1	2	8.3	140	-20	+10.5	+12	T1
CL6-BG101-PG	25	15	28.8	100.0*	J, K, M	1	10	31.4	50	-28	+13	+15	T2
CL6-BG181-PG	25	15	28.8	180.0*	J, K, M	2	18	54.3	32	-48	+13	+15	T3
CL6-BH080-PG	30	20	34.5	8.0	J, K, M	1	2	4.5	275	-16	+8	+12	T1
CL6-BH150-PG	30	20	34.5	15.0*	J, K, M	1	2	9.1	175	-20	+10.5	+12	T1
CL6-BH400-PG	30	20	34.5	40.0	J, K, M	1	5	12.2	65	-24	+10.5	+12	T2
CL6-BH680-PG	30	20	34.5	68.0*	J, K, M	1	8	31.0	60	-24	+13	+15	T2
CL6-BH101-PG	30	20	34.5	100.0	J, K, M	2	12	19.0	40	-28	+10.5	+12	T3
CL6-BH151-PG	30	20	34.5	150.0*	J, K, M	2	18	46.0	35	-48	+13	+15	T3

See footnotes at end of table.

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TABLE I. Capacitor characteristics (cont).

Type designation 1/	Voltage			Capacitance 2/	Cap. tolerance	DC leakage (max)		Dissipation factor (max)	Impedance (max)	Capacitance change			Case size
	Rated (+85°C)	Derated (+125°C)	Surge (+85°C)			+25°C	+85°C & +125°C			-55°C	+85°C	+125°C	
	volts, dc	volts, dc	volts, dc	µF		µA	µA	percent	ohms	percent	percent	percent	
CL6-BJ050-PG	50	30	57.5	5.0	J, K, M	1	2	3.4	400	-16	+5	+6	T1
CL6-BJ100-PG	50	30	57.5	10.0*	J, K, M	1	2	6.0	250	-24	+8	+9	T1
CL6-BJ250-PG	50	30	57.5	25.0	J, K, M	1	5	11.2	95	-20	+10.5	+12	T2
CL6-BJ470-PG	50	30	57.5	47.0*	J, K, M	1	9	21.4	70	-28	+13	+15	T2
CL6-BJ600-PG	50	30	57.5	60.0	J, K, M	2	12	13.6	45	-16	+10.5	+12	T3
CL6-BJ820-PG	50	30	57.5	82.0*	J, K, M	2	16	24.9	45	-32	+13	+15	T3
CL6-BK040-PG	60	40	69.0	4.0	J, K, M	1	2	3.0	550	-16	+5	+6	T1
CL6-BK8R2-PG	60	40	69.0	8.2*	J, K, M	1	2	5.0	275	-24	+8	+9	T1
CL6-BK200-PG	60	40	69.0	20.0	J, K, M	1	5	7.6	105	-16	+10.5	+12	T2
CL6-BK390-PG	60	40	69.0	39.0*	J, K, M	1	9	20.7	90	-28	+10.5	+12	T2
CL6-BK500-PG	60	40	69.0	50.0	J, K, M	2	12	15.3	50	-16	+10.5	+12	T3
CL6-BK680-PG	60	40	69.0	68.0*	J, K, M	2	16	20.7	50	-32	+10.5	+12	T3
CL6-BL3R5-PG	75	50	86.2	3.5	J, K, M	1	2	2.5	650	-16	+5	+6	T1
CL6-BL6R8-PG	75	50	86.2	6.8*	J, K, M	1	2	4.1	300	-20	+8	+9	T1
CL6-BL150-PG	75	50	86.2	15.0	J, K, M	1	5	7.5	150	-16	+8	+9	T2
CL6-BL330-PG	75	50	86.2	33.0*	J, K, M	1	10	17.5	90	-24	+10.5	+15	T2
CL6-BL400-PG	75	50	86.2	40.0	J, K, M	2	12	15.2	60	-16	+10.5	+12	T3
CL6-BL560-PG	75	50	86.2	56.0*	J, K, M	2	17	26.0	60	-28	+10.5	+15	T3
CL6-BN2R5-PG	100	65	115.0	2.5	J, K, M	1	2	5.0	950	-16	+7	+8	T1
CL6-BN4R7-PG	100	65	115.0	4.7*	J, K, M	1	2	3.6	500	-16	+7	+8	T1
CL6-BN110-PG	100	65	115.0	11.0	J, K, M	1	4	5.0	200	-16	+7	+8	T2
CL6-BN220-PG	100	65	115.0	22.0*	J, K, M	1	9	11.8	100	-16	+7	+8	T2
CL6-BN300-PG	100	65	115.0	30.0	J, K, M	2	12	9.1	80	-16	+7	+8	T3
CL6-BN430-PG	100	65	115.0	43.0*	J, K, M	2	17	19.7	70	-20	+7	+8	T3
CL6-BP1R7-PG	125	85	144.0	1.7	J, K, M	1	2	7.0	1,250	-16	+7	+8	T1
CL6-BP3R6-PG	125	85	144.0	3.6*	J, K, M	1	2	4.1	600	-16	+7	+8	T1
CL6-BP090-PG	125	85	144.0	9.0	J, K, M	1	5	10.2	240	-16	+7	+8	T2
CL6-BP140-PG	125	85	144.0	14.0*	J, K, M	1	7	12.7	167	-16	+7	+8	T2
CL6-BP250-PG	125	85	144.0	25.0*	J, K, M	2	13	19.0	93	-16	+7	+8	T3

1/ Complete type designation will include additional symbols to indicate style and capacitance tolerance.
2/ The asterisk (*) indicates high-capacitance series.

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TABLE II. Cross-reference data.

FROM Type designation 1/ MIL-C-3965/24A	TO Part number 2/ MIL-C-39006/9B	FROM Type designation 1/ MIL-C-3965/24A	TO Part number 2/ MIL-C-39006/9B
	M39006/09-		M39006/09-
CL6-BB300MPG	4001	CL6-BE150MPG	4055
CL6-BB300KPG	4002	CL6-BE150KPG	4056
CL6-BB300JPG	4003	CL6-BE150JPG	4057
CL6-BB680MPG	4004	CL6-BE330MPG	4058
CL6-BB680KPG	4005	CL6-BE330KPG	4059
CL6-BB680JPG	4006	CL6-BE330JPG	4060
CL6-BB141MPG	4007	CL6-BE700MPG	4061
CL6-BB141KPG	4008	CL6-BE700KPG	4062
CL6-BB141JPG	4009	CL6-BE700JPG	4063
CL6-BB271MPG	4010	CL6-BE121MPG	4064
CL6-BB271KPG	4011	CL6-BE121KPG	4065
CL6-BB271JPG	4012	CL6-BE121JPG	4066
CL6-BB331MPG	4013	CL6-BE171MPG	4067
CL6-BB331KPG	4014	CL6-BE171KPG	4068
CL6-BB331JPG	4015	CL6-BE171JPG	4069
CL6-BB561MPG	4016	CL6-BE271MPG	4070
CL6-BB561KPG	4017	CL6-BE271KPG	4071
CL6-BB561JPG	4018	CL6-BE271JPG	4072
CL6-BC250MPG	4021	CL6-BG100MPG	4075
CL6-BC250KPG	4022	CL6-BG100KPG	4076
CL6-BC250JPG	4023	CL6-BG100JPG	4077
CL6-BC560MPG	4024	CL6-BG220MPG	4078
CL6-BC560KPG	4025	CL6-BG220KPG	4079
CL6-BC560JPG	4026	CL6-BG220JPG	4080
CL6-BC221MPG	4027	CL6-BG101MPG	4081
CL6-BC221KPG	4028	CL6-BG101KPG	4082
CL6-BC221JPG	4029	CL6-BG101JPG	4083
CL6-BC431MPG	4030	CL6-BG181MPG	4084
CL6-BC431KPG	4031	CL6-BG181KPG	4085
CL6-BC431JPG	4032	CL6-BG181JPG	4086
CL6-BD200MPG	4035	CL6-BH080MPG	4089
CL6-BD200KPG	4036	CL6-BH080KPG	4090
CL6-BD200JPG	4037	CL6-BH080JPG	4091
CL6-BD470MPG	4038	CL6-BH150MPG	4092
CL6-BD470KPG	4039	CL6-BH150KPG	4093
CL6-BD470JPG	4040	CL6-BH150JPG	4094
CL6-BD101MPG	4041	CL6-BH400MPG	4095
CL6-BD101KPG	4042	CL6-BH400KPG	4096
CL6-BD101JPG	4043	CL6-BH400JPG	4097
CL6-BD181MPG	4044	CL6-BH680MPG	4098
CL6-BD181KPG	4045	CL6-BH680KPG	4099
CL6-BD181JPG	4046	CL6-BH680JPG	4100
CL6-BD251MPG	4047	CL6-BH101MPG	4101
CL6-BD251KPG	4048	CL6-BH101KPG	4102
CL6-BD251JPG	4049	CL6-BH101JPG	4103
CL6-BD391MPG	4050	CL6-BH151MPG	4104
CL6-BD391KPG	4051	CL6-BH151KPG	4105
CL6-BD391JPG	4052	CL6-BH151JPG	4106

See footnotes at end of table.

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TABLE II. Cross-reference data. -Continued

FROM	TO	FROM	TO
Type designation 1/ MIL-C-3965/24A	Part number 2/ MIL-C-39006/9B	Type designation 1/ MIL-C-3965/24A	Part number 2/ MIL-C-39006/9B
	M39006/09-		M39006/09-
CL6-BJ050MPG	4109	CL6-BL150JPG	4157
CL6-BJ050KPG	4110	CL6-BL330MPG	4158
CL6-BJ050JPG	4111	CL6-BL330KPG	4159
CL6-BJ100MPG	4112	CL6-BL330JPG	4160
CL6-BJ100KPG	4113	CL6-BL400MPG	4161
CL6-BJ100JPG	4114	CL6-BL400KPG	4162
CL6-BJ250MPG	4115	CL6-BL400JPG	4163
CL6-BJ250KPG	4116	CL6-BL560MPG	4164
CL6-BJ250JPG	4117	CL6-BL560KPG	4165
CL6-BJ470MPG	4118	CL6-BL560JPG	4166
CL6-BJ470KPG	4119	CL6-BN2R5MPG	4169
CL6-BJ470JPG	4120	CL6-BN2R5KPG	4170
CL6-BJ600MPG	4121	CL6-BN2R5JPG	4171
CL6-BJ600KPG	4122	CL6-BN4R7MPG	4172
CL6-BJ600JPG	4123	CL6-BN4R7KPG	4173
CL6-BJ820MPG	4124	CL6-BN4R7JPG	4174
CL6-BJ820KPG	4125	CL6-BN110MPG	4175
CL6-BJ820JPG	4126	CL6-BN110KPG	4176
CL6-BK040MPG	4129	CL6-BN110JPG	4177
CL6-BK040KPG	4130	CL6-BN220MPG	4178
CL6-BK040JPG	4131	CL6-BN220KPG	4179
CL6-BK8R2MPG	4132	CL6-BN220JPG	4180
CL6-BK8R2KPG	4133	CL6-BN300MPG	4181
CL6-BK8R2JPG	4134	CL6-BN300KPG	4182
CL6-BK200MPG	4135	CL6-BN300JPG	4183
CL6-BK200KPG	4136	CL6-BN430MPG	4184
CL6-BK200JPG	4137	CL6-BN430KPG	4185
CL6-BK390MPG	4138	CL6-BN430JPG	4186
CL6-BK390KPG	4139	CL6-BP1R7MPG	4189
CL6-BK390JPG	4140	CL6-BP1R7KPG	4190
CL6-BK500MPG	4141	CL6-BP1R7JPG	4191
CL6-BK500KPG	4142	CL6-BP3R6MPG	4192
CL6-BK500JPG	4143	CL6-BP3R6KPG	4193
CL6-BK680MPG	4144	CL6-BP3R6JPG	4194
CL6-BK680KPG	4145	CL6-BP090MPG	4195
CL6-BK680JPG	4146	CL6-BP090KPG	4196
CL6-BL3R5MPG	4149	CL6-BP090JPG	4197
CL6-BL3R5KPG	4150	CL6-BP140MPG	4198
CL6-BL3R5JPG	4151	CL6-BP140KPG	4199
CL6-BL6R8MPG	4152	CL6-BP140JPG	4200
CL6-BL6R8KPG	4153	CL6-BP250MPG	4201
CL6-BL6R8JPG	4154	CL6-BP250KPG	4202
CL6-BL150MPG	4155	CL6-BP250JPG	4203
CL6-BL150KPG	4156		

1/ Complete type designation will include additional symbol to indicate style.
2/ Part number referenced will replace styles CL66 and CL67.

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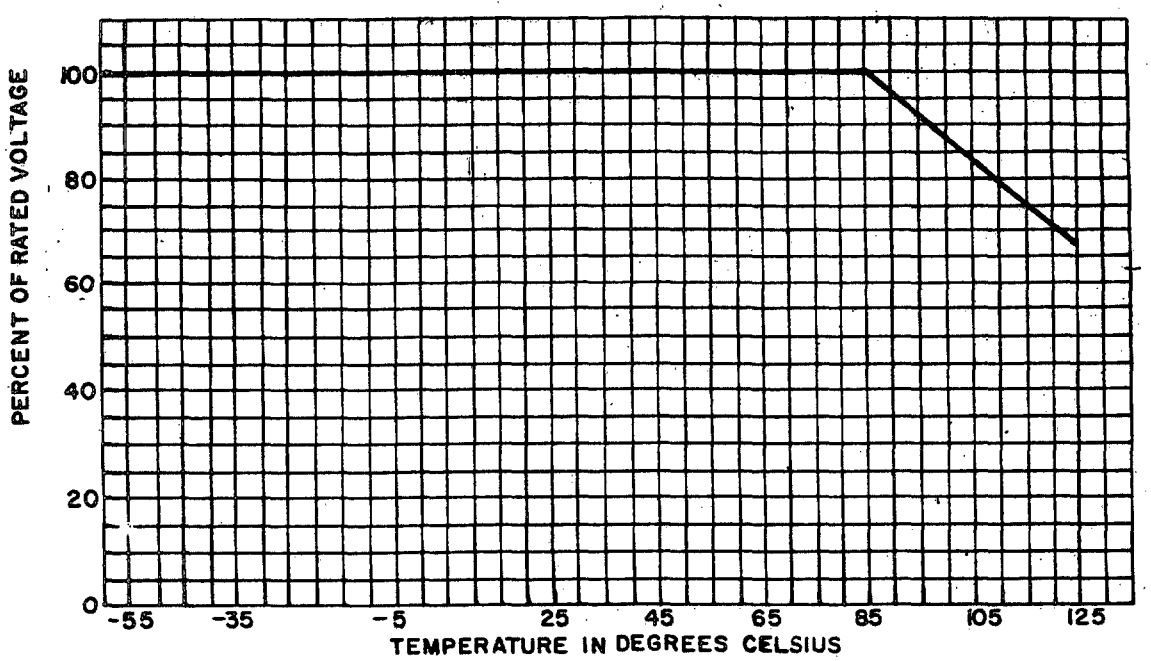


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Tubular; uninsulated (CL66), insulated (CL67).
Seal type - Hermetic (symbol G).
Terminals - Axial wire-lead.
Rated temperature - -55° C to +85° C, voltage derated to ±125° C (see figure 2).
Rated voltage - See table I.
Capacitance (cap) (nom) - See table I.
Capacitance tolerance - Symbol J (±5%), K (±10%) or M (±20%).

DC leakage (DCL) (max) (at +25° C):
See table I.

Dissipation factor (DF) (max):
See table I.

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - 2, 000 vdc.
Terminal to case - Not applicable.

Insulation resistance:
Sleeving to case (min): 100 megohms.
Terminal to case - Not applicable.

Temperature cycling:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - 115% of table I value.

Solderability: In accordance with MIL-C-3965.

Terminal strength:
Pull test - In accordance with MIL-C-3965.
Bend test - Not applicable.
Torque test - Not applicable.
Wire-lead bend test - In accordance with MIL-C-3965.
Twist test - Not applicable.

Surge voltage: See table I.

Moisture resistance:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±8% of initial measurement.
DF (max) - 115% of table I value.

Low temperature (storage):
DCL (max) - See table I (25° C).
ΔCap -Within ±5% of initial measurement.
DF (max) - See table I.

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Stability at low and high temperature:
Step 1 (+25° C):
DCL (max) - See table I.
Cap - Within tolerance of table I value.
DF (max) - See table I.
Step 2 (-55° C):
Impedance - See table I.
ΔCap (max) - See table I.
Step 3 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of step 1 value.
DF (max) - See table I.
Step 4 (+85° C):
DCL (max) - See table I.
ΔCap (max) - See table I.
DF (max) - See table I.
Step 5 (+125° C):
DCL (max) - See table I.
ΔCap (max) - See table I.
DF (max) - See table I.
Step 6 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - See table I.

Reverse voltage: Not applicable.

Life at +85° C:
2,000-hr (qualification):
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - See table I (25° C).
240-hr (group B):
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - See table I (25° C).
1,760-hr (group C):
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - See table I (25° C).

Barometric pressure:
Test voltage - Rated voltage.

Life at +125° C:
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - See table I (25° C).
DWV (min) (CL67, only) - 2,000 vdc.
IR (min) (CL67, only) - 100 megohms.

Marking:
Polarity - Series of plus (+) symbols encircling positive end (see figure 1).

NOTE: The capacitors covered by this specification sheet are substitutable for those covered by MIL-C-3965/24. See table II.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MU
Navy - EC, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Air Force - 80

Agent:
DSA - ES

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